

Maclaurin's Theorem:

The power series expansion of a function $f(x)$ is

$f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots + a_nx^n + \dots$ where $a_0, a_1, a_2, \dots, a_n, \dots$ are constants and x is a variable. Now we find all constants by finding successive derivatives of the power series and evaluating them at $x = 0$

$$f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots + a_nx^n + \dots \longrightarrow \text{(i)}$$

$$f(0) = a_0 \longrightarrow \text{(ii)}$$

$$f'(x) = a_1 + 2a_2x + 3a_3x^2 + \dots + na_nx^{n-1} + \dots$$

$$f'(0) = a_1 \longrightarrow \text{(iii)}$$

$$f''(x) = 2a_2 + 6a_3x + \dots + n(n-1)a_nx^{n-2} + \dots$$

$$f''(0) = 2a_2, \quad a_2 = \frac{1}{2!} f''(0) \longrightarrow \text{(iv)}$$

$$f'''(x) = 6a_3 + \dots + n(n-1)(n-2)a_nx^{n-3} + \dots$$

$$f'''(0) = 6a_3, \quad a_3 = \frac{1}{3!} f'''(0) \longrightarrow \text{(v)}$$

Putting (ii), (iii), (iv), (v) in (i)

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots + \frac{x^n}{n!} f^n(0) + \dots$$

This expansion is called **Maclaurin series** or **Maclaurin's theorem**

Note:

A function f can be expanded in the Maclaurin series if the function is defined in the interval containing 0 and its derivatives exist at $x = 0$. The expansion is valid only if it is convergent.

Taylor's Series Expansions of Functions:

If f is defined in the interval containing 'a' and its derivatives of all orders exist at $x = a$ then we can expand $f(x)$ as

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \dots + \frac{f^n(a)}{n!}(x-a)^n + \dots$$

Proof:

Let

$$f(x) = a_0 + a_1(x-a) + a_2(x-a)^2 + a_3(x-a)^3 + a_4(x-a)^4 + \dots + a_n(x-a)^n + \dots \longrightarrow \text{(i)}$$

$$f(a) = a_0 \longrightarrow \text{(ii)}$$

$$f'(x) = a_1 + 2a_2(x-a) + 3a_3(x-a)^2 + 4a_4(x-a)^3 + \dots + na_n(x-a)^{n-1} + \dots$$

$$f'(a) = a_1 \longrightarrow \text{(iii)}$$

$$f''(x) = 0 + 2a_2 + 6a_3(x-a) + 12a_4(x-a)^2 + \dots + n(n-1)a_n(x-a)^{n-2} + \dots$$

$$f''(a) = 2a_2$$

$$\frac{1}{2} f''(a) = a_2 \longrightarrow \text{(iv)}$$

$$f'''(x) = 0 + 6a_3 + 24a_4(x-a) + \dots + n(n-1)(n-2)a_n(x-a)^{n-3} + \dots$$

$$f'''(a) = 6a_3$$

$$\frac{1}{6} f'''(a) = a_3$$

$$\frac{1}{3!} f'''(a) = a_3 \longrightarrow \text{(v)}$$

Similarly

$$\frac{1}{n!} f^n(a) = a_n \longrightarrow \text{(vi)}$$

Putting (ii), (iii), (iv), (v), (vi) in (i)

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots + \frac{f^n(a)}{n!}(x-a)^n + \dots$$

This expansion of function $f(x)$ is called **Taylor series** expansion.

EXERCISE 2.8**Q.1 Apply the Maclaurin series expansion to prove that:**

(i) $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$

Solution:

Let

$$f(x) = \ln(1+x) \quad \text{and} \quad f(0) = 0$$

$$f'(x) = \frac{1}{1+x} \quad \text{and} \quad f'(0) = 1$$

$$f''(x) = \frac{-1}{(1+x)^2} \quad \text{and} \quad f''(0) = -1$$

$$f'''(x) = \frac{2}{(1+x)^3} \quad \text{and} \quad f'''(0) = 2$$

$$f^{iv}(x) = \frac{-6}{(1+x)^4} \quad \text{and} \quad f^{iv}(0) = -6$$

Applying Maclaurin's series expansion,

$$f(x) = f(0) + \frac{f'(0)x}{1} + \frac{f''(0)x^2}{2} + \frac{f'''(0)x^3}{3} + \dots$$

$$\ln(1+x) = 0 + \frac{1}{1}x + \frac{-1}{2}x^2 + \frac{2}{3}x^3 + \dots$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

(ii) **Prove that** $\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{4} - \frac{x^6}{6} + \dots$

Solution:

$$\text{Let } f(x) = \cos x \quad \text{and} \quad f(0) = 1$$

$$f'(x) = -\sin x \quad \text{and} \quad f'(0) = 0$$

$$f''(x) = -\cos x \quad \text{and} \quad f''(0) = -1$$

$$f'''(x) = \sin x \quad \text{and} \quad f'''(0) = 0$$

$$f^{iv}(x) = \cos x \quad \text{and} \quad f^{iv}(0) = 1$$

$$f^v(x) = -\sin x \quad \text{and} \quad f^v(0) = 0$$

$$f^{vi}(x) = -\cos x \quad \text{and} \quad f^{vi}(0) = -1$$

Applying Maclaurin's series expansion,

$$f(x) = f(0) + \frac{f'(0)}{1}x + \frac{f''(0)}{2}x^2 + \frac{f'''(0)}{3}x^3 + \dots$$

$$\cos x = 1 + \frac{0}{1}x + \frac{-1}{2}x^2 + \frac{0}{3}x^3 + \frac{1}{4}x^4 + \dots$$

$$\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{4} - \frac{x^6}{6} + \dots$$

$$(iii) \quad \sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16} + \dots$$

Solution:

Let

$$f(x) = \sqrt{1+x} \quad \text{and} \quad f(0) = 1$$

$$f'(x) = \frac{1}{2\sqrt{1+x}} \quad \text{and} \quad f'(0) = \frac{1}{2}$$

$$f''(x) = \frac{-1}{4(1+x)^{\frac{3}{2}}} \quad \text{and} \quad f''(0) = \frac{-1}{4}$$

$$f'''(x) = \frac{3}{8(1+x)^{\frac{5}{2}}} \quad \text{and} \quad f'''(0) = \frac{3}{8}$$

$$f^{iv}(x) = \frac{-15}{16(1+x)^{\frac{7}{2}}} \quad \text{and} \quad f^{iv}(0) = \frac{-15}{16}$$

Applying Maclaurin's series expansion,

$$f(x) = f(0) + \frac{f'(0)}{1}x + \frac{f''(0)}{2}x^2 + \frac{f'''(0)}{3}x^3 + \dots$$

$$\sqrt{1+x} = 1 + \frac{1}{2}x + \frac{\frac{-1}{4}}{2}x^2 + \frac{\frac{3}{8}}{3}x^3 + \dots$$

$$\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16} + \dots$$

$$(iv) \quad e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \dots$$

Solution:

Let

$$f(x) = e^x \quad \text{and} \quad f(0) = 1$$

$$f'(x) = e^x \quad \text{and} \quad f'(0) = 1$$

$$f''(x) = e^x \quad \text{and} \quad f''(0) = 1$$

$$f'''(x) = e^x \quad \text{and} \quad f'''(0) = 1$$

Applying Maclaurin's series expansion,

$$f(x) = f(0) + \frac{f'(0)}{1}x + \frac{f''(0)}{2}x^2 + \frac{f'''(0)}{3}x^3 + \dots$$

$$e^x = 1 + \frac{1}{1}x + \frac{1}{2}x^2 + \frac{1}{3}x^3 + \dots$$

$$\Rightarrow e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \dots$$

$$(v) \quad e^{2x} = 1 + 2x + \frac{4x^2}{2} + \frac{8x^3}{3} + \dots$$

Solution:

Let

$$f(x) = e^{2x} \quad \text{and} \quad f(0) = 1$$

$$f'(x) = 2e^{2x} \quad \text{and} \quad f'(0) = 2$$

$$f''(x) = 4e^{2x} \quad \text{and} \quad f''(0) = 4$$

$$f'''(x) = 8e^{2x} \quad \text{and} \quad f'''(0) = 8$$

$$f^{iv}(x) = 16e^{2x} \quad \text{and} \quad f^{iv}(0) = 16$$

Applying Maclaurin's series expansion,

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2}x^2 + \frac{f'''(0)}{3}x^3 + \dots$$

$$e^{2x} = 1 + 2x + \frac{4x^2}{2} + \frac{8x^3}{3} + \dots$$

Q.2 Show that $\cos(x+h) = \cos x - h \sin x - \frac{h^2}{2} \cos x + \frac{h^3}{3} \sin x + \dots$

and evaluate $\cos 61^\circ$.

Solution:

Let

$$f(x+h) = \cos(x+h)$$

$$\Rightarrow f(x) = \cos x$$

$$f'(x) = -\sin x \quad \text{and} \quad f''(x) = -\cos x$$

$$f'''(x) = \sin x \quad \text{and} \quad f^{iv}(x) = \cos x$$

Applying Taylor's Theorem, we have

$$f(x+h) = f(x) + \frac{h}{1}f'(x) + \frac{h^2}{2}f''(x) + \frac{h^3}{3}f'''(x) + \dots$$

$$\cos(x+h) = \cos x - h \sin x + \frac{h^2}{2} \cos x - \frac{h^3}{3} \sin x + \dots$$

For $x = 60^\circ$ and $h = 1^\circ$ or $h = 0.017455$

$$\cos(60^\circ + 1^\circ) = \cos 60^\circ - (0.017455) \sin 60^\circ - \frac{(0.017455)^2}{2} \cos 60^\circ + \frac{(0.017455)^3}{3} \sin 60^\circ$$

$$\begin{aligned}
 \cos 61^\circ &= \frac{1}{2} - \frac{\sqrt{3}}{2}(0.017455) - \frac{1}{4}(0.017455)^2 + \frac{\sqrt{3}}{12}(0.017455)^3 \\
 &\approx 0.5 - 0.015116 - 0.0000761 + 0.00000076 \\
 &\approx 0.4848
 \end{aligned}$$

Q.3 Show that $2^{x+h} = 2^x \left[1 + (\ln 2)h + \frac{(\ln 2)^2}{\underline{2}}h^2 + \frac{(\ln 2)^3}{\underline{3}}h^3 + \dots \right]$

Solution:

$$\text{Let } f(x+h) = 2^{x+h}$$

$$\Rightarrow f(x) = 2^x$$

$$f'(x) = 2^x (\ln 2) \quad \text{and} \quad f''(x) = 2^x (\ln 2)^2$$

$$f'''(x) = 2^x (\ln 2)^3 \quad \text{and} \quad f^{iv}(x) = 2^x (\ln 2)^4$$

Applying Taylor's Theorem

$$f(x+h) = f(x) + \frac{h}{\underline{1}} f'(x) + \frac{h^2}{\underline{2}} f''(x) + \frac{h^3}{\underline{3}} f'''(x) + \dots$$

$$2^{x+h} = 2^x + \frac{h}{\underline{1}} \cdot 2^x (\ln 2) + \frac{h^2}{\underline{2}} 2^x (\ln 2)^2 + \dots$$

$$2^{x+h} = 2^x \left[1 + (\ln 2)h + (\ln 2)^2 \frac{h^2}{\underline{2}} + (\ln 2)^3 \frac{h^3}{\underline{3}} + \dots \right]$$